



Answers & Solutions:

1. C. $7x + 46$

$$\begin{aligned} j(x) &= 7x^2 - 3 \\ (j \circ h)(x) &= j(h(x)) \\ &= 7h(x)^2 - 3 \\ &= 7(\sqrt{x+7})^2 - 3 \\ &= 7(x+7) - 3 \\ &= 7x + 49 - 3 \\ &= 7x + 46 \end{aligned}$$

2. D. 12

$$\begin{aligned} x &= \text{number of students who like both Calculus and Geometry} \\ 27 - x &= \text{number of students who like Calculus only} \\ 25 - x &= \text{number of students who like Geometry only} \\ 40 &= \text{total number of students in the class} \\ 40 &= x + (27 - x) + (25 - x) = 52 - x \\ x &= 52 - 40 = 12 \end{aligned}$$

3. A. An element of S is a set with 1, 2, 3, 4, or 5 elements.

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4. A. B

$$\begin{aligned} A \cup B &= \{x, y\} \cup \{x, y, z\} \\ &= \{x, x, y, y, z\} \\ &= \{x, y, z\} \end{aligned}$$

5. B. $\{(a,1), (a,2), (a,3), (b,1), (b,2), (b,3)\}$

$$\begin{aligned} A &= \{1, 2, 3\} \\ B &= \{a, b\} \\ B \times A &= \left\{ (a,1), (a,2), (a,3), \right. \\ &\quad \left. (b,1), (b,2), (b,3) \right\} \end{aligned}$$

6. C. Set

7. B. $\{1, 3, 5, 7, 9\}$

8. C. $\{(1, a), (2, a), (1, b), (2, b)\}$

$$\begin{aligned} A &= \{1, 2\} \\ B &= \{a, b\} \\ A \times B &= \{(1, a), (1, b), (2, a), (2, b)\} \end{aligned}$$

9. C. $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$ *

The two sets, $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$, are equal.

10. A. infinite

11. B. $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

$$\begin{aligned} S &= \{0^2, 1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2\} \\ &= \{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\} \end{aligned}$$

12. B. $\{1, 2, 5, 6\}$

$$\begin{aligned} \{1, 2, 5\} \cup \{1, 2, 6\} &= \{1, 1, 2, 2, 5, 6\} \\ &= \{1, 2, 5, 6\} \end{aligned}$$

13. A. $\{1, 2\}$

$$\{1, 2, 5\} \cap \{1, 2, 6\} = \{1, 2\}$$

14. C. intersection

15. C. $\{3\}$

$$\begin{aligned} \{1, 2, 3\} \setminus \{1, 2, 5\} &= \{1, 2, 3\} - \{1, 2, 5\} \\ &= \{3\} \end{aligned}$$

16. A. 2

$A = 10$ (number of people who likes apples)
 $3 =$ number of people who likes apples, bananas and carrots
 $2 =$ number of people who likes apples and bananas
 $3 =$ number of people who likes apples and carrots
 $X =$ number of people who likes apples only

$$\begin{aligned} 3 + 2 + 3 + X &= A \\ 8 + X &= 10 \\ X &= 2 \end{aligned}$$

17. D. 10

Number of elements of set A, $n(A) = 20$
Number of elements of set B, $n(B) = 20$
Number of elements of the union of set A and set B, $n(A \cup B) = 40$
Number of elements of the intersection of set A and set B, $n(A \cap B) = ?$

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ 40 &= 20 + 20 - n(A \cap B) \\ 40 &= 40 - n(A \cap B) \\ n(A \cap B) &= 10 \end{aligned}$$

18. No answer (insufficient information)

19. A. Z contains both X and Y

20. C. $n(A \cup B \cup C) = 90$

The number of elements of the union of Set A, Set B and Set C is equal to 90. $n(A \cup B \cup C) = 90$

21. B. 3.5

$$\begin{aligned} f(x) &= -2x + 8 \\ x &= -2f^{-1}(x) + 8 \\ 2f^{-1}(x) &= 8 - x \\ f^{-1}(x) &= \frac{8-x}{2} \\ f^{-1}(1) &= \frac{8-1}{2} = \frac{7}{2} \text{ or } 3.5 \end{aligned}$$

22. C. $\frac{1}{3} \sin^{-1}\left(\frac{2x-1}{x}\right)$

$$\begin{aligned} f(x) &= \frac{1}{2 - \sin 3x} \\ x &= \frac{1}{2 - \sin 3f^{-1}(x)} \\ 2 - \sin 3f^{-1}(x) &= \frac{1}{x} \\ \sin 3f^{-1}(x) &= 2 - \frac{1}{x} \\ 3f^{-1}(x) &= \sin^{-1}\left(2 - \frac{1}{x}\right) \\ f^{-1}(x) &= \frac{1}{3} \sin^{-1}\left(\frac{2x-1}{x}\right) \end{aligned}$$



23. B. 4

$$\text{mean} = \frac{5+4+6+1}{4} = 4$$

24. C. 5

Median = the middle term of the arrangement of the terms from lowest to highest.

The middle terms are 2 and 8. The median is the average of the 2 middle terms.

$$\text{median} = \frac{2+8}{2} = 5$$

25. B. 3, 14, 19, 25, 14

{3, 14, 19, 25, 14} (Answer)

Arrange from lowest to highest: {3, 14, 14, 19, 25}

$$\text{mean} = \frac{3+14+14+19+25}{5} = 15$$

$$\text{range} = \text{highest term} - \text{lowest term} = 25 - 3 = 22$$

median = 14 (the middle term)

26. D. 5

Arrange from lowest to highest: {3, 3, 4, 5, 7, 8, 11, 15}

$$\text{mean} = \frac{3+3+4+5+7+8+11+15}{8} = 7$$

Median: average of the two middle terms

$$\text{median} = \frac{5+7}{2} = 6$$

Mode: the most frequently occurring term

mode = 3

5 (Answer)

27. C. 90

Mode: the most frequently occurring term

mode = 90 (Answer)

28. B. 80

Arrange from lowest to highest: {60, 60, 70, 75, 80, 80, 85, 85, 90, 90, 90}
Since there are 11 terms, the 6th term is the median.

median = 80 (Answer)

29. C. 30

$$\text{Range} = \text{highest term} - \text{lowest term} = 90 - 60 = 30$$

30. A. More students received a 94 than any other score.

31. D. 2

Mode: the most frequently occurring term

mode = 2 (Answer)

32. D. 6

$$\text{Mean} = \frac{1+2+2+8+9+14}{6} = 6$$

33. D. 5.5

Median: average of the two middle terms

$$\text{median} = \frac{5+6}{2} = 5.5$$

34. B. 20

Arrange from lowest to highest: {3, 7, 17, 19, 20, 21, 31, 43, 46}

The median is the middle term which is the 5th term.

median = 20

35. A. 300

Arrange from lowest to highest: {29, 100, 200, 300, 450, 1029, 2001}

The median is the middle term which is the 4th term.

median = 300

36. B. 9

Mode: the most frequently occurring term

mode = 9 (Answer)

37. D. 90

Mode: the most frequently occurring term

mode = 90 (Answer)

38. C. 17

Mode: the most frequently occurring term

mode = 17 (Answer)

39. B. 444

$$\text{Mean} = \frac{100+1050+320+600+150}{5} = 444$$

40. A. 11

$$\text{Mean} = \frac{3+6+27+13+6+8+12+20+5+10}{10} = 11$$

41. D. 30

$$\text{Mean} = \frac{62+18+39+13+16+37+25}{7} = 30$$

42. B. 0

$$f(0) = (0)\sec(0) = 0$$

43. C. x

$$g(x) = \tan(x)$$

$$(g \circ f)(x) = g(f(x))$$

$$= \tan(f(x))$$

$$= \tan(\tan^{-1} x)$$

$$= x$$

44. D. sqrt of (x^2 - 4x + 5)

$$g(x) = \sqrt{x^2 + 1}$$

$$(g \circ f)(x) = g(f(x))$$

$$= \sqrt{f(x)^2 + 1}$$

$$= \sqrt{(x-2)^2 + 1}$$

$$= \sqrt{x^2 - 4x + 4 + 1}$$

$$= \sqrt{x^2 - 4x + 5}$$



45. D. sqrt of ($x^2 + 8$)

$$\begin{aligned} f(x) &= \sqrt{x+4} \\ f(x^2 + 4) &= \sqrt{x^2 + 4 + 4} \\ &= \sqrt{x^2 + 8} \end{aligned}$$

The lower quartile is the average of the 2nd and the 3rd term.

$$\begin{aligned} \text{lower quartile} &= 25\text{th percentile} \\ &= \frac{11+12}{2} \\ &= 11.5\% \end{aligned}$$

46. D. sqrt of 2

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \sqrt{2 \cos\left(2\pi \sin \frac{\pi}{2}\right)} \\ &= \sqrt{2} \end{aligned}$$

END

47. C. 50th

48. D. Averages of position

49. B. 3%

5th decile = 50th percentile

Arrange from lowest to highest: {2%, 2%, 2.09%, 2.7%, 3%, 3%, 4%, 7%, 7.2%, 7.5%, 8%}

k = 50 (50th percentile)

n = 11 (total number of terms)

$$\begin{aligned} i &= \frac{k}{100}(n+1) \\ &= \frac{50}{100}(11+1) \\ &= 6 \end{aligned}$$

5th decile is the 6th term which is 3%.
 (Answer)

50. C. 11.5%

The lower quartile is the 25th percentile.

Arrange from lowest to highest:
 {10%, 11%, 12%, 14%, 15%, 16%, 17%, 19%, 21%, 23%}

k = 25 (25th percentile)

n = 10 (total number of terms)

$$\begin{aligned} i &= \frac{k}{100}(n+1) \\ &= \frac{25}{100}(10+1) \\ &= 2.75 \end{aligned}$$